Instructions

Part 1: Configure a Home Wireless Router.

You are installing a new home wireless router at a friend's house. You will need to change settings on the router to enhance security and meet your friend's requirements.

Step 1: Change DHCP settings.

a. Open the Home Wireless Router GUI and change the router IP and DHCP settings according to the information in the Addressing Table.

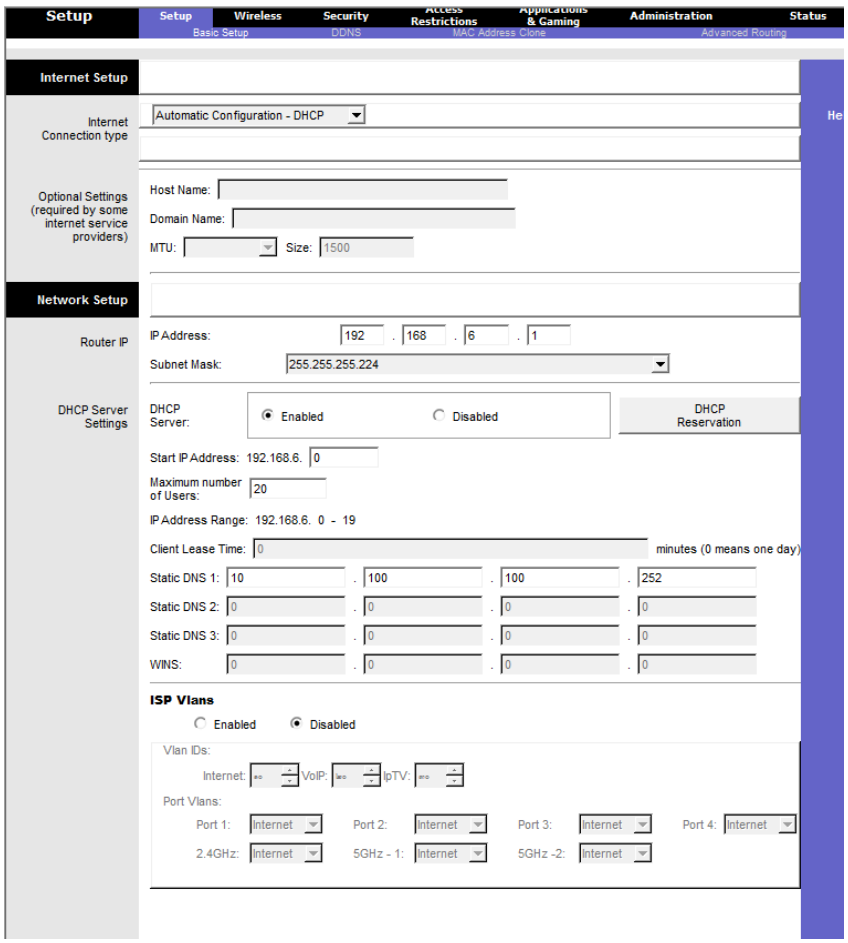
b. Permit a maximum of **20** addresses to be issued by the router.

c. Configure the DHCP server to start with IP address .**3** of the LAN network.

d. Configure the internet interface of the router to receive its IP address over DHCP.

Verify the address. What address did it receive?

e. Configure the static DNS server to the address in the Addressing Table.

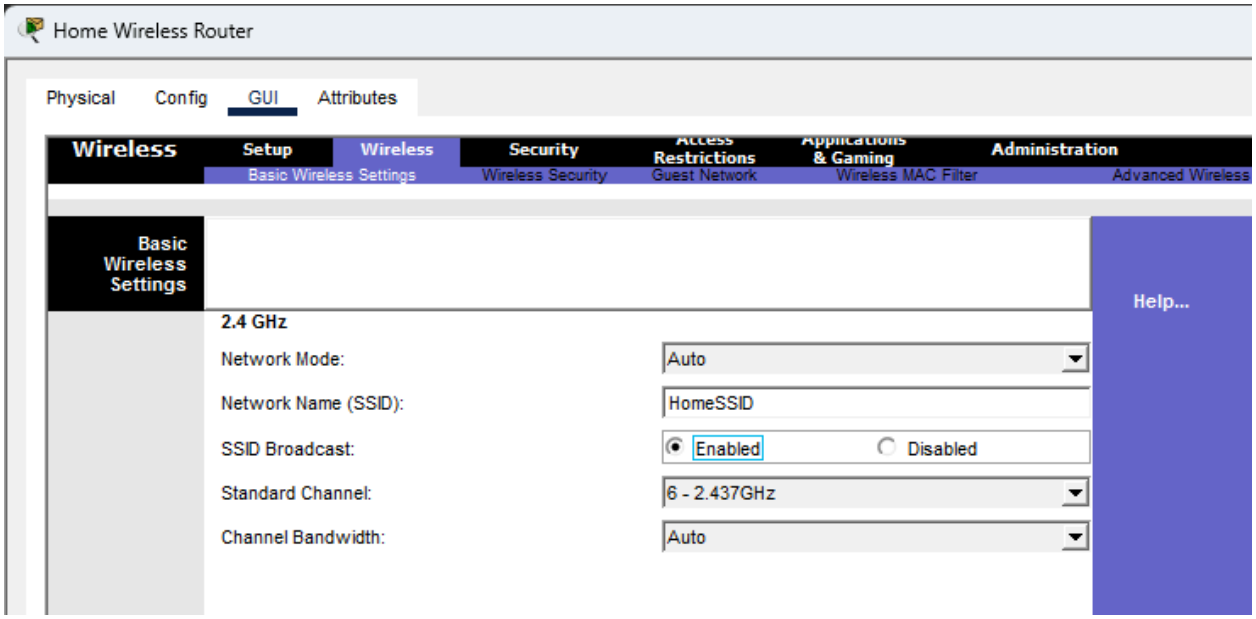


Step 2: Configure the Wireless LAN.

a. The network will use the 2.4GHz Wireless LAN interface. Configure the interface with the SSID shown in the Wireless LAN information table.

b. Use **channel 6**.

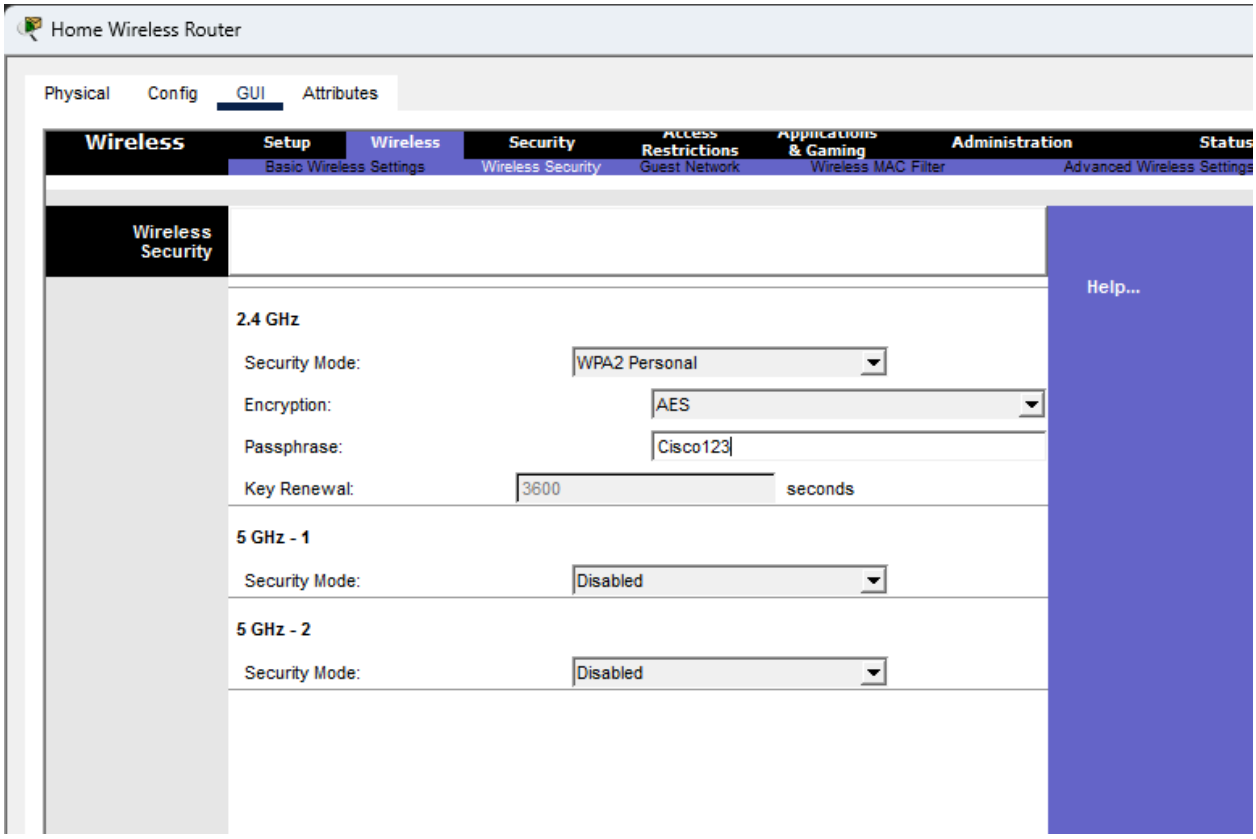
c. Be sure that all wireless hosts in the home will be able to see the SSID.

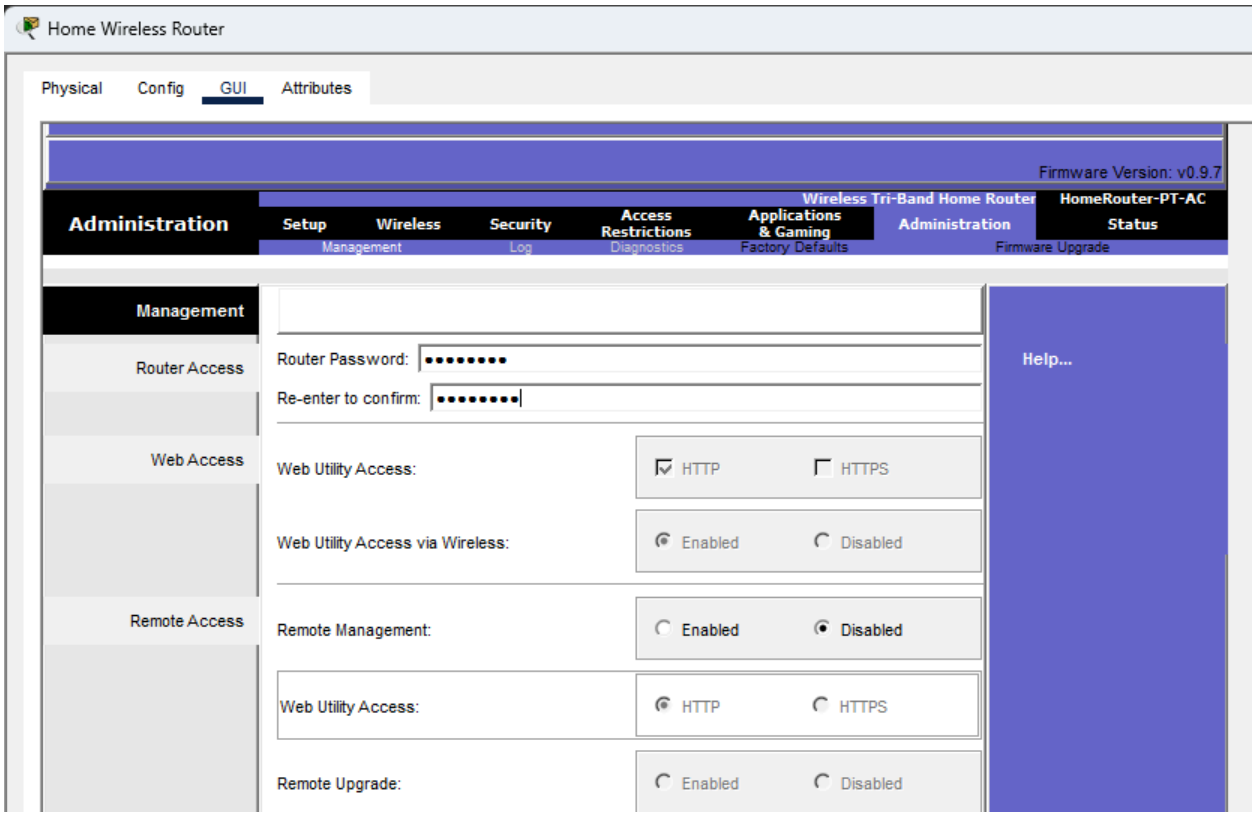


Step 3: Configure security.

a. Configure wireless LAN security. Use WPA2 Personal and the passphrase shown in the Wireless LAN information table.

b. Secure the router by changing the default password to the value shown in the Wireless LAN information table.


Home Wireless Router
Physical
Config
GUI
Attributes
Wireless
Setup
Wireless
Security
Access Restrictions
Applications & Gaming
Administration
Status
Basic Wireless Settings
Wireless Security
Guest Network
Wireless MAC Filter
Advanced Wireless Settings
Wireless Security
Help...
2.4 GHz
Security Mode:
WPA2 Personal
Encryption:
AES
Passphrase:
Cisco123
Key Renewal:
3600
seconds
5 GHz - 1
Security Mode:
Disabled
5 GHz - 2
Security Mode:
Disabled


Home Wireless Router
Physical
Config
GUI
Attributes
Firmware Version: v0.9.7
Wireless Tri-Band Home Router
HomeRouter-PT-AC
Administration
Setup
Wireless
Security
Access Restrictions
Applications & Gaming
Administration
Status
Management
Log
Diagnostics
Factory Defaults
Firmware Upgrade
Management
Router Access
Web Access
Remote Access
Help...
Router Password:
Re-enter to confirm:
Web Utility Access:
HTTP
HTTPS
Web Utility Access via Wireless:
Enabled
Disabled
Remote Management:
Enabled
Disabled
Web Utility Access:
HTTP
HTTPS
Remote Upgrade:
Enabled
Disabled

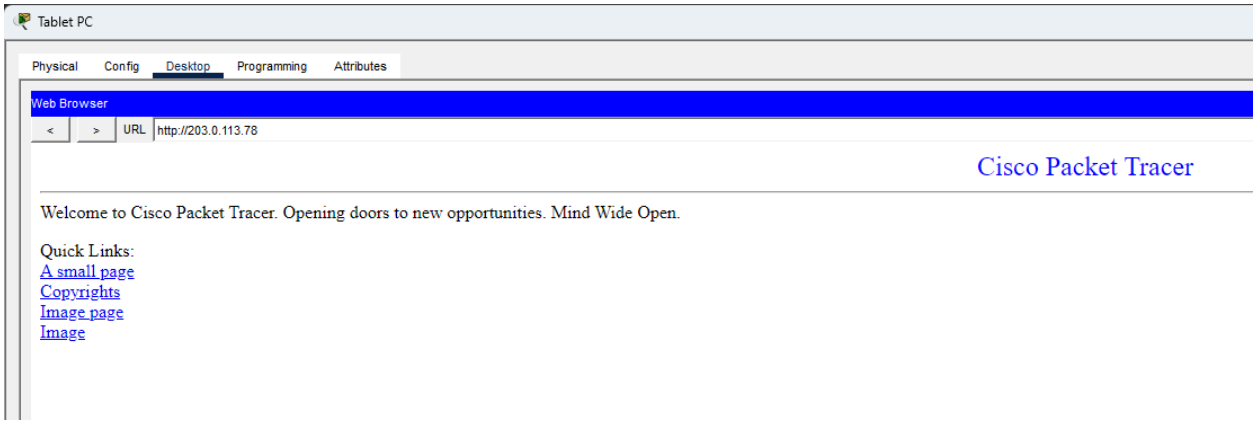
Step 4: Connect clients to the network.

a. Open the PC Wireless app on the desktop of the laptop and configure the client to connect to the network.

b. Open the Config tab on the Tablet PC and Smartphone and configure the wireless interfaces to connect to the wireless network.

c. Verify connectivity. The hosts should be able to ping each other and the web server. They should also be able to reach the web server URL.





Part 2: Configure a WLC Controller Network

Configure the wireless LAN controller with two WLANs. One WLAN will use WPA2-PSK authentication. The other WLAN will use WPA2-Enterprise authentication. You will also configure the WLC to use an SNMP server and configure a DHCP scope that will be used by the wireless management network.

Step 1: Configure VLAN interfaces.

a. From the Enterprise Admin, navigate to the WLC-1 management interface via a web browser. To log into WLC-1, use **admin** as the username and **Cisco123** as the password.

b. Configure an interface for the first WLAN.

Name: **WLAN 2**

VLAN Identifier: **2**

Port Number: **1**

Interface IP Address: **192.168.2.254**

Netmask: **255.255.255.0**

Gateway: **RTR-1 G0/0/0.2 address**

Primary DHCP Server: **Gateway address**

c. Configure an interface for the second WLAN.

Name: **WLAN 5**

VLAN Identifier: **5**

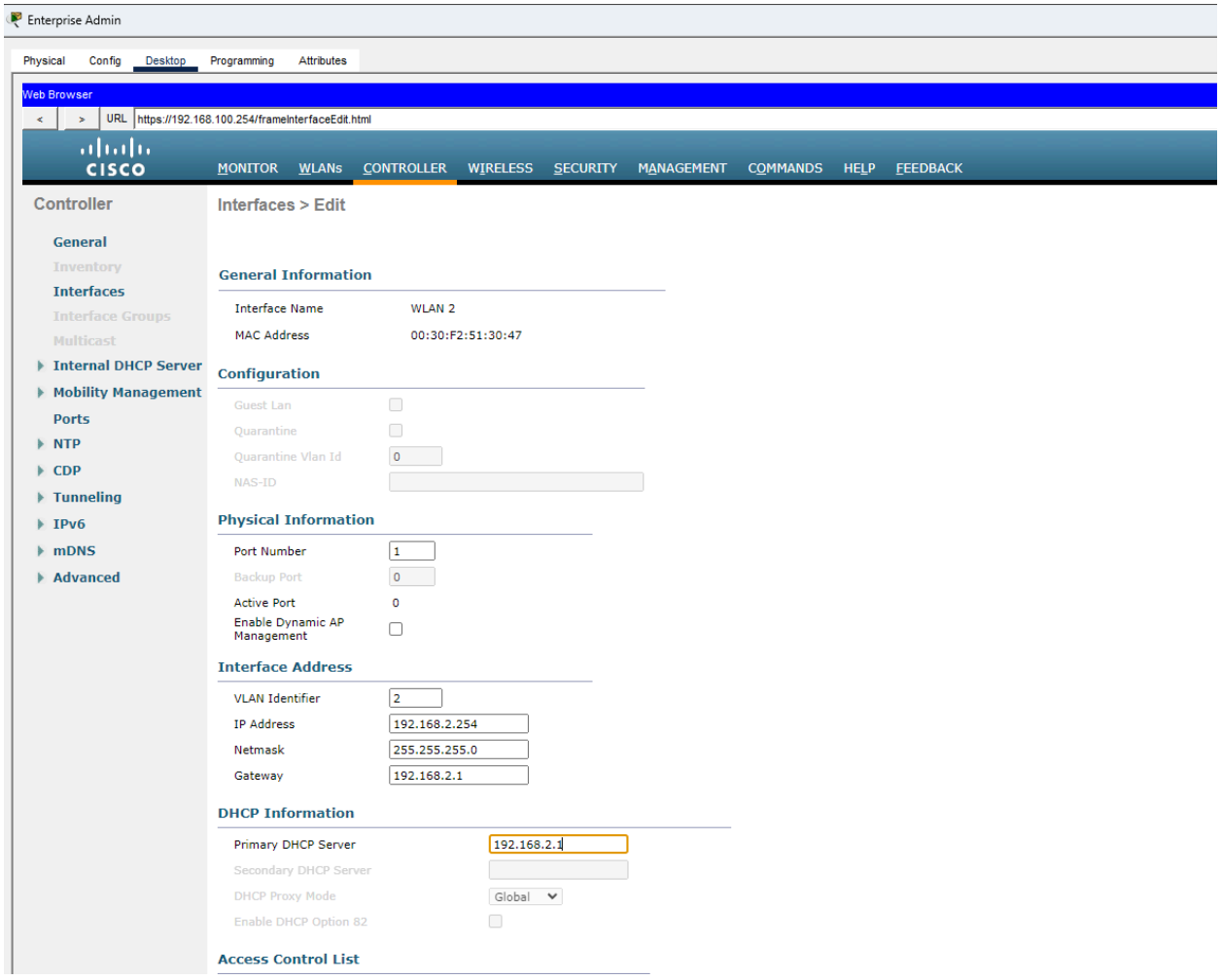
Port Number: **1**

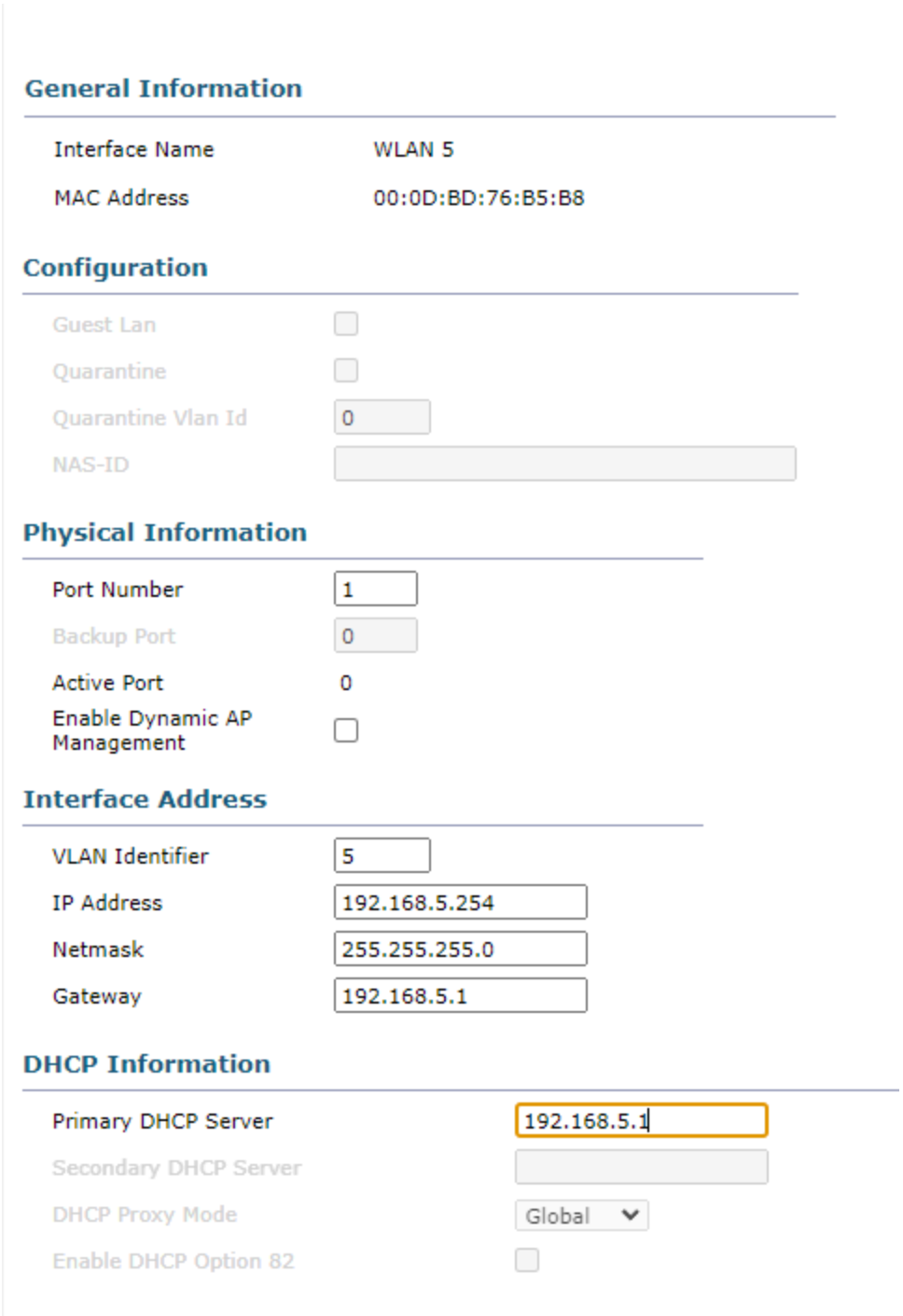
Interface IP Address: **192.168.5.254**

Netmask: **255.255.255.0**

Gateway: **RTR-1 interface G0/0/0.5 address**

Primary DHCP Server: **Gateway address**


Enterprise Admin
Physical
Config
Desktop
Programming
Attributes
Web Browser
URL
https://192.168.100.254/frameInterfaceEdit.html
MONITOR
WLANs
CONTROLLER
WIRELESS
SECURITY
MANAGEMENT
COMMANDS
HELP
FEEDBACK
Controller
General
Inventory
Interfaces
Interface Groups
Multicast
Internal DHCP Server
Mobility Management
Ports
NTP
CDP
Tunneling
IPv6
mDNS
Advanced
Interfaces > Edit
General Information
Interface Name
WLAN 2
MAC Address
00:30:F2:51:30:47
Configuration
Guest Lan
Quarantine
Quarantine Vlan Id
0
NAS-ID
Physical Information
Port Number
1
Backup Port
0
Active Port
0
Enable Dynamic AP Management
Interface Address
VLAN Identifier
2
IP Address
192.168.2.254
Netmask
255.255.255.0
Gateway
192.168.2.1
DHCP Information
Primary DHCP Server
192.168.2.1
Secondary DHCP Server
DHCP Proxy Mode
Global
Enable DHCP Option 82
Access Control List



Step 2: Configure a DHCP scope for the wireless management network.

Configure and enable an internal DHCP scope as follows:

Scope Name: **management**

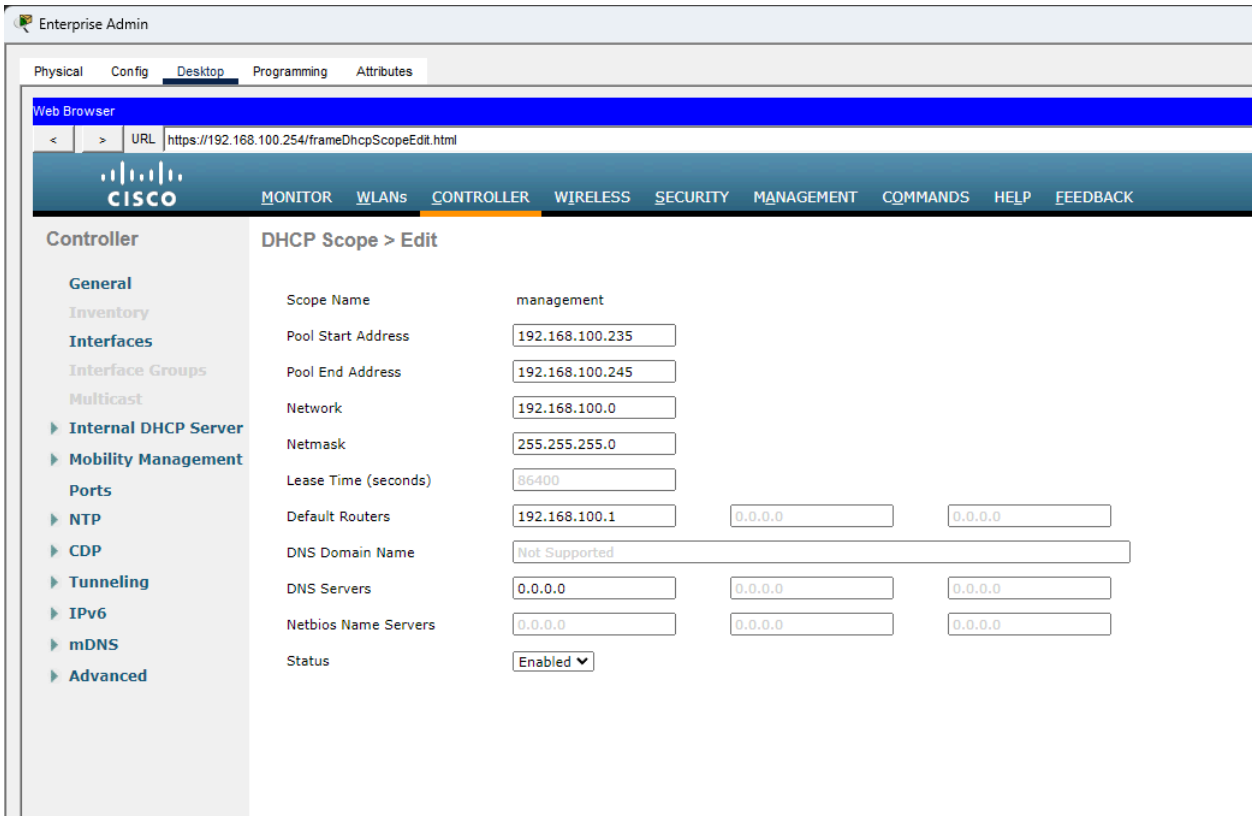
Pool Start Address: **192.168.100.235**

Pool End Address: **192.168.100.245**

Network: **192.168.100.0**

Netmask: **255.255.255.0**

Default Routers: **192.168.100.1**



Step 3: Configure the WLC with external server addresses.

a. Configure the RADIUS server information as follows:

Sever Index: **1**

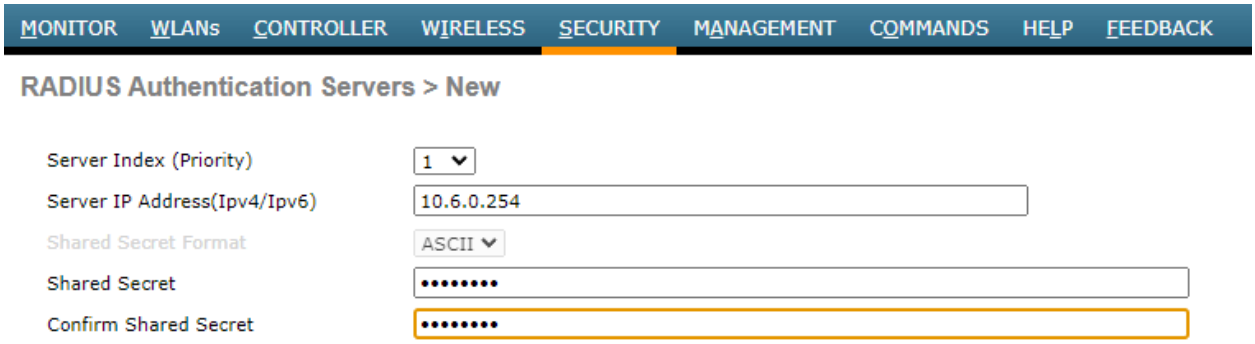
Sever Address: **10.6.0.254**

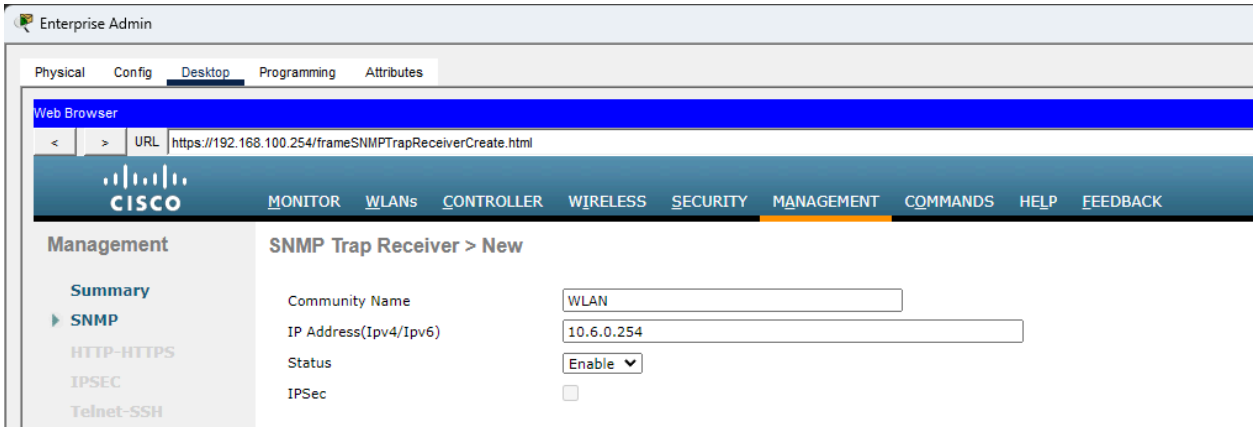
Shared Secret: **RadiusPW**

b. Configure the WLC to send logs information to an SNMP server.

Community Name: **WLAN**

IP Address: **10.6.0.254**





Step 4: Create the WLANs.

a. Create the first WLAN:

Profile Name: **Wireless VLAN 2**

WLAN SSID: **SSID-2**

ID: **2**

Interface: **WLAN 2**

Security: **WPA2-PSK**

Passphrase: **Cisco123**

Under the Advanced tab, go to the FlexConnect section. Enable **FlexConnect Local Switching** and **FlexConnect Local Auth**.

b. Create the second WLAN:

Profile Name: **Wireless VLAN 5**

WLAN SSID: **SSID-5**

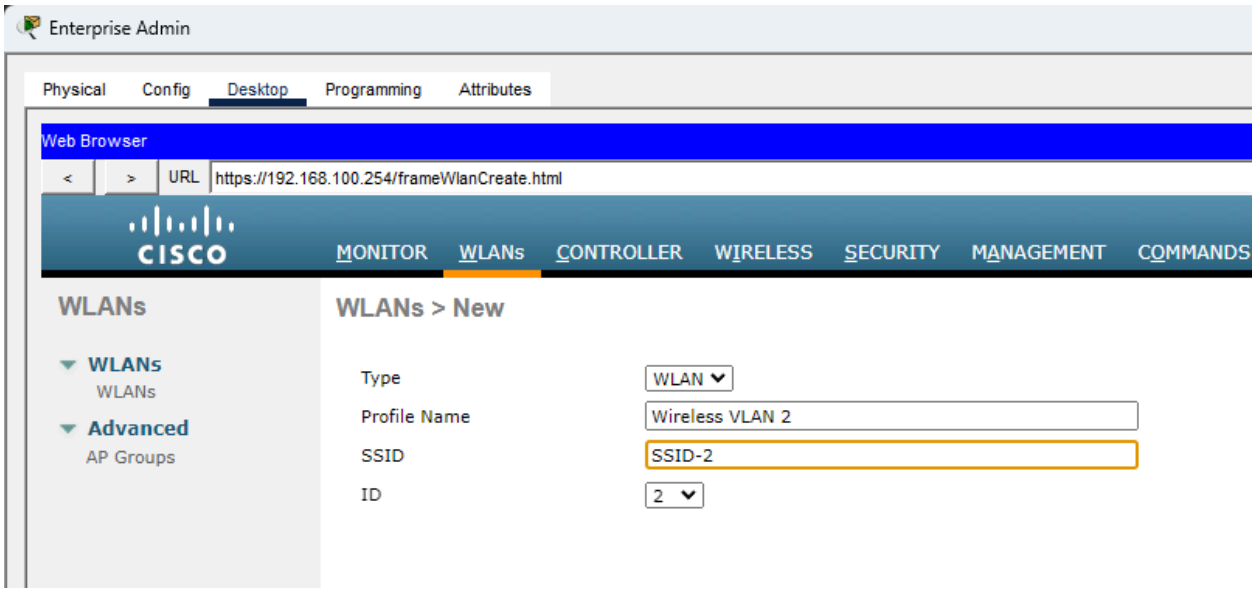
Interface: **WLAN 5**

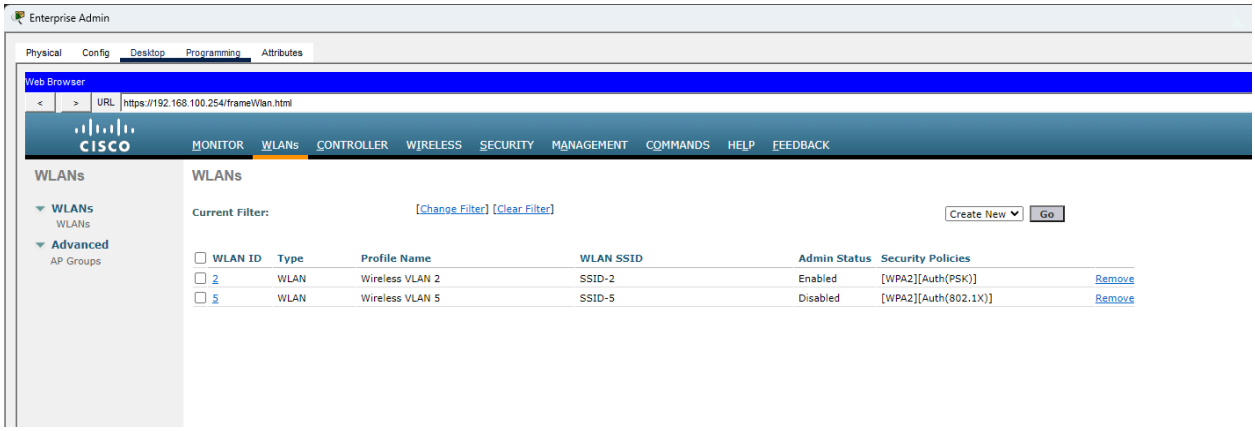
ID: **5**

Security: **802.1x - WPA2-Enterprise**

Configure the WLAN to use the RADIUS server for authentication.

Make the **FlexConnect** settings as was done in Step 4a.





Step 5: Configure the hosts to connect to the WLANs.

Use the desktop PC Wireless app to configure the hosts as follows:

a. Wireless Host 1 should connect to Wireless VLAN 2.

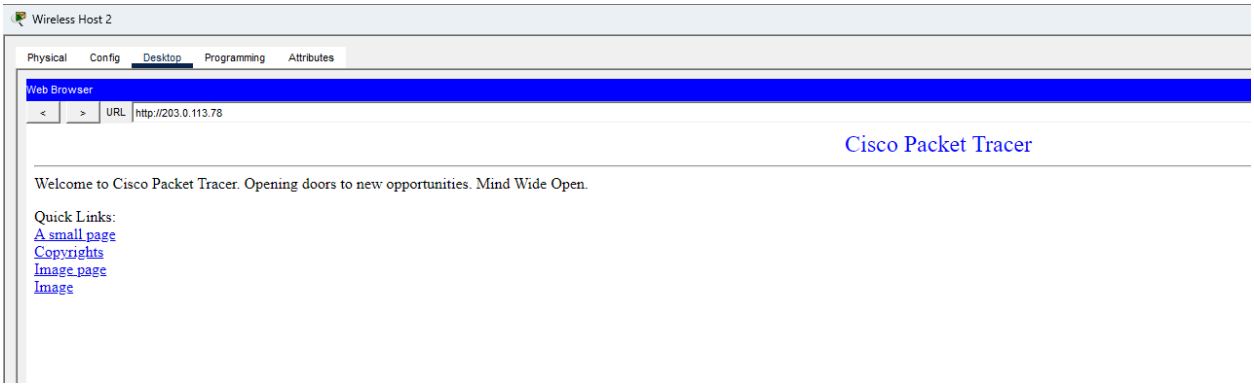
b. Wireless Host 2 should connect to Wireless VLAN 5 using the credentials in the WLAN information table.

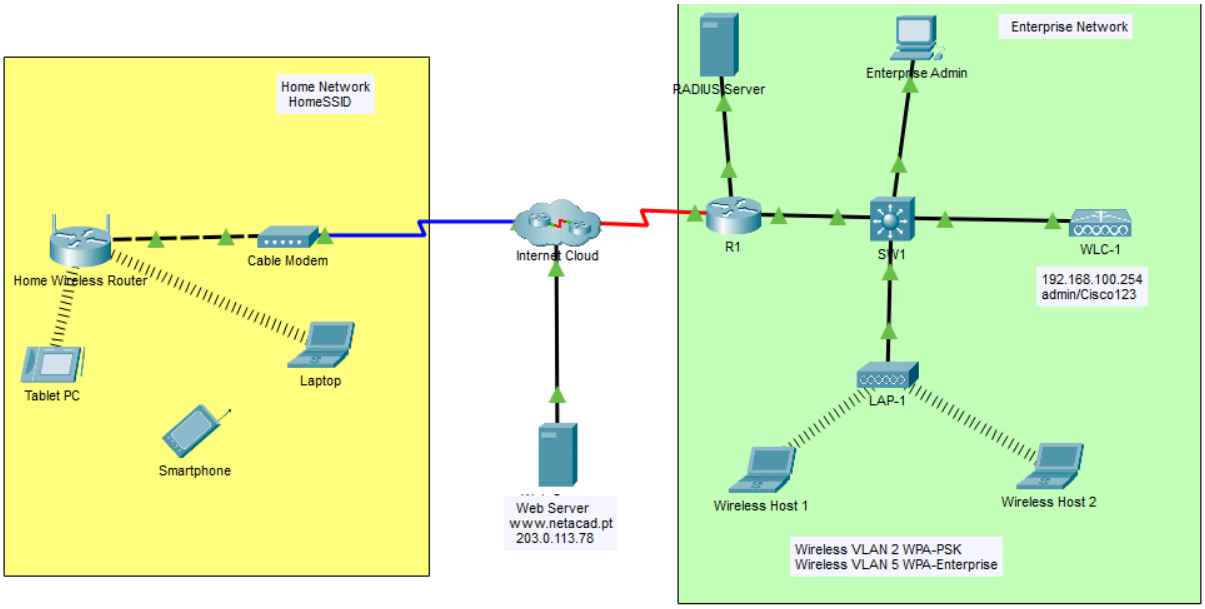

Wireless Host 1
Physical
Config
Desktop
Programming
Attributes
Link Information
Connect
Profiles
Below is a list of available wireless networks. To search for more wireless networks, click the Refresh button. To view more information about a network, select the wireless network name. To connect to that network, click the Connect button below.
Wireless Network Name
CH
Signal
SSID-2
1
100%
Site Information
Wireless Mode Infrastructure
Network Type Mixed B/G/N
Radio Band Auto
Security WPA2-PSK
MAC Address 0060.5C8A.B002
Refresh
Connect
2.4GHz
Adapter is Active
Wireless-N Notebook Adapter
Wireless Network Monitor v1.0
Model No. WPC300N



Step 6: Test connectivity.

Test connectivity between the wireless hosts and the Web Server by ping and URL.




Home Network
HomeSSID
Home Wireless Router
Cable Modem
Tablet PC
Laptop
Smartphone
Internet Cloud
Web Server
www.netacad.pt
203.0.113.78
Enterprise Network
RADIUS Server
Enterprise Admin
R1
SW1
WLC-1
192.168.100.254
admin/Cisco123
LAP-1
Wireless Host 1
Wireless Host 2
Wireless VLAN 2 WPA-PSK
Wireless VLAN 5 WPA-Enterprise